Question 2 : Counting Bank Accounts by Salary Category | conditional

aggregation (Medium)

(REF : https://leetcode.com/problems/count-salary-categories/description/)

Table: Accounts

```
+-------------------+--------+
| Column Name | Type |
+-------------------+---------+
| account_id     |    int    |
| income          |    int    |
+-------------------+---------+
```

account_id is the primary key (column with unique values) for this table.
Each row contains information about the monthly income for one bank account.

Write a solution to calculate the number of bank accounts for each salary category. The salary categories are:
"Low Salary": All the salaries are strictly less than $20000.
"Average Salary": All the salaries in the inclusive range [$20000, $50000].
"High Salary": All the salaries strictly greater than $50000.
The result table must contain all three categories. If there are no accounts in a category, return 0.
Return the result table in any order.
The result format is in the following example.

Example 1:
Input:
Accounts table:

```
+----------------+-----------+
| account_id | income |
+----------------+-----------+
| 3               | 108939 |
| 2               | 12747    |
| 8               | 87709    |
| 6               | 91796    |
+----------------+------------+
```

Output:

```
+----------------+---------------------+
| category     | accounts_count |
+----------------+---------------------+
| Low Salary |         1               |
| Average Salary |    0              |
| High Salary |         3              |
+----------------+----------------------+
```

Explanation:
Low Salary: Account 2.
Average Salary: No accounts.
High Salary: Accounts 3, 6, and 8.

-- Insert statements to satisfy all conditions
INSERT INTO Accounts (account_id, income) VALUES (1, 15000); -- Low Salary
INSERT INTO Accounts (account_id, income) VALUES (2, 25000); -- Average Salary
INSERT INTO Accounts (account_id, income) VALUES (3, 35000); -- Average Salary
INSERT INTO Accounts (account_id, income) VALUES (4, 48000); -- Average Salary
INSERT INTO Accounts (account_id, income) VALUES (5, 52000); -- High Salary
INSERT INTO Accounts (account_id, income) VALUES (6, 75000); -- High Salary
INSERT INTO Accounts (account_id, income) VALUES (7, 60000); -- High Salary
INSERT INTO Accounts (account_id, income) VALUES (8, 19000); -- Low Salary

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-- question 2

Accounts table:

account_id	income	category
3	108939	high
2	12747	low
8	87709	high
6	91796	high

Output:

category	accounts_count
Low Salary	1
Average Salary	0
High Salary	3

< 20000 Low Salary
>= 20000 <= 50000 Average Salary
> 50000 High Salary

```
select 'Low Salary' as category, SUM(CASE WHEN INCOME < 20000 THEN 1 ELSE 0 END)
AS accounts_count
from accounts
union all
select 'Average Salary', SUM(CASE WHEN INCOME >= 20000 and INCOME <=50000 THEN 1 ELSE 0
END)
from accounts
union all
select 'High Salary', SUM(CASE WHEN INCOME > 50000 THEN 1 ELSE 0
END)
from accounts;
```

```
select 'Low Salary' as category, SUM(INCOME < 20000)
AS accounts_count
from accounts
union all
select 'Average Salary', SUM(INCOME >= 20000 and INCOME <=50000)
from accounts
union all
select 'High Salary', SUM(INCOME > 50000)
from accounts;
```

===============

```
| CATEGORY |
| --------------- |
| High Salary |
| Low Salary  |
| High Salary |
| High Salary |
```

```
SELECT category, count(*) as accounts_count
FROM
(SELECT CASE
WHEN INCOME < 20000 THEN 'Low Salary'
WHEN INCOME >=20000 AND INCOME <=50000 THEN 'Average Salary'
ELSE 'High Salary'
END
AS CATEGORY FROM ACCOUNTS) d group by category;
```

Output:

expected

category	accounts_count
Low Salary	1
Average Salary	0
High Salary	3

our output

CATEGORY	accounts_count
High Salary	3
Low Salary	1

```
with temp as
(select 'High Salary' as category
union all
select 'Low Salary'
union all
select 'Average Salary'
)
select * from temp;
```

temp left side

category
High Salary
Low Salary
Average Salary

right side

CATEGORY	accounts_count
High Salary	3
Low Salary	1

left outer join

```
with temp as
(select 'High Salary' as category
union all
select 'Low Salary'
union all
select 'Average Salary'
)
```

```sql
SELECT temp.category, coalesce(accounts_count,0) as accounts_count
from temp left join
(SELECT category, count(*) as accounts_count
FROM
(SELECT CASE
WHEN INCOME < 20000 THEN 'Low Salary'
WHEN INCOME >=20000 AND INCOME <=50000 THEN 'Average Salary'
ELSE 'High Salary'
END
AS CATEGORY FROM ACCOUNTS) d group by category) e
on temp.category = e.category;
```

==========

```sql
with temp1 as
(select 'High Salary' as category
union all
select 'Low Salary'
union all
select 'Average Salary'
),
temp2 as (SELECT category, count(*) as accounts_count
FROM
(SELECT CASE
WHEN INCOME < 20000 THEN 'Low Salary'
WHEN INCOME >=20000 AND INCOME <=50000 THEN 'Average Salary'
ELSE 'High Salary'
END
AS CATEGORY FROM ACCOUNTS) d group by category)

SELECT temp1.category, coalesce(accounts_count,0) as accounts_count
from temp1 left join temp2
on temp1.category = temp2.category;
```